

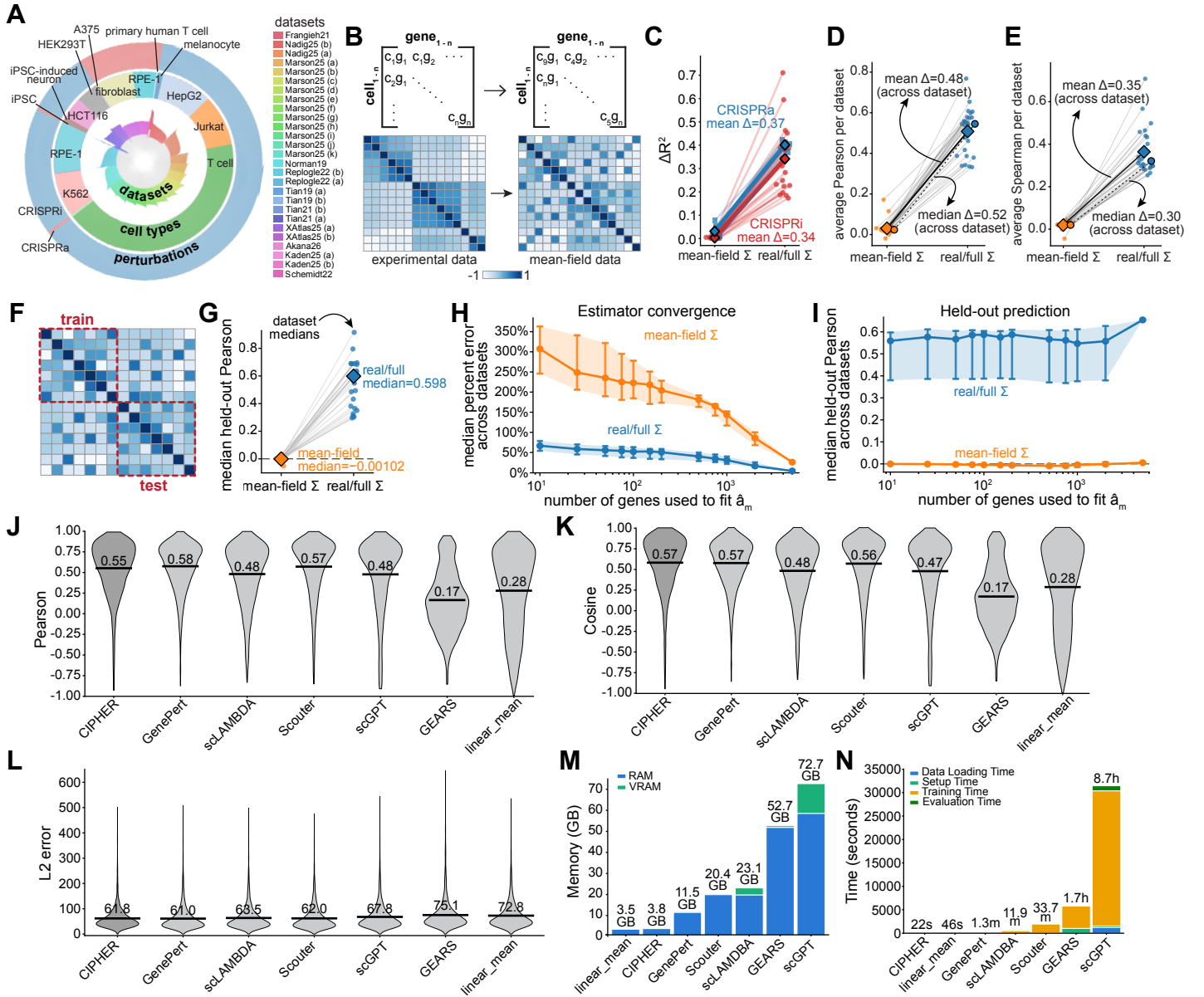


Fig. 3: The forward problem: to what extent can covariance structure explain response to known gene perturbations from Perturb-seq.

a, We consider 27 Perturb-seq datasets, comprising CRISPRi (21 datasets) and CRISPRa (6 datasets) gene perturbations (outermost ring) and a variety of different cell types (middle ring). The innermost ring shows the log-number of control (gray) and perturbed (colored) samples for each perturbation in each dataset. **b**, Experimental (real) and ‘mean-field’ (shuffled over cells, not genes) gene expression count matrices, and corresponding covariance matrices. **c**, Improvement in transcriptome-wide prediction accuracy obtained by using the experimentally measured covariance matrix instead of the mean-field covariance. Each line corresponds to one dataset and connects the performance using the mean-field covariance (left) to that obtained using the real covariance matrix (right). The vertical axis shows the increase in coefficient of determination (ΔR^2). Red points correspond to CRISPRi datasets and blue points to CRISPRa datasets, with mean improvements indicated for each perturbation modality. Large diamonds and circles denote the mean and median across datasets. **d**, Average Pearson correlation between predicted and observed perturbation responses for each dataset using the mean-field covariance (left) and the real covariance (right). Each connected pair represents one dataset. Large diamonds denote the mean across datasets, while annotations indicate the mean and median improvements. **e**, Same analysis as in **(d)**, but using Spearman correlation as the performance metric. Individual datasets are connected by gray lines, with mean values shown as diamonds and overall mean and median improvements annotated. **f**, Schematic of the train-test evaluation used to assess generalization across included genes. A block of the covariance matrix including the perturbed gene is estimated, while prediction accuracy is evaluated on held-out test genes (lower-right block), as indicated by the dashed partition. **g**, Median held-out Pearson correlation obtained using covariance matrices estimated from the training subset consisting of 50% of genes selected randomly. Each point corresponds to one dataset, comparing predictions using the mean-field covariance and the real covariance. Dataset medians are indicated by diamonds. **h**, Dependence of fitting parameter error $\hat{a}_m$ on the number of genes used to estimate it. The horizontal axis denotes the number of genes included during estimation, while the vertical axis shows the median percent error across datasets. Blue curves correspond to real covariance matrices and orange curves correspond to mean-field covariance estimates. Error bars denote variability across datasets. **i**, Held-out prediction performance as a function of the number of genes used to estimate the response fitting parameter $\hat{a}_m$. Median Pearson correlation across datasets is plotted for real covariance matrices (blue) and mean-field covariance matrices (orange), with error bars indicating variation across datasets. **j**, Distribution of Pearson correlations between predicted and experimentally observed perturbation responses for CIPHER and other state-of-the-art methods. Violin plots summarize the distribution across perturbations, horizontal bars indicate medians and median values are labeled numerically above each distribution. **k**, Same comparison as in **(j)**, using cosine similarity between predicted and observed perturbation responses. Median values are indicated for each method. **l**, Distribution of transcriptome-wide L2 prediction error for each method. Violin plots summarize the distribution across perturbations, with horizontal bars indicating median errors and numerical median values shown above each violin. **m**, Peak memory requirements during model execution. Blue bars indicate peak RAM usage and green bars indicate peak GPU memory (VRAM) usage for each method. Memory consumption is reported in gigabytes (GB). **n**, Runtime comparison across methods. Stacked bars partition total execution time into data loading (blue), setup (green), training (orange), and evaluation components (dark green). Total runtimes are annotated above each bar, with times reported in seconds, minutes, or hours as appropriate.